

Problem 2.13

Suppose that an accumulation function $a(t)$ is differentiable and satisfies the property

$$a(s+t) = a(s) \cdot a(t)$$

for all non-negative real numbers s and t .

1. Using the definition of derivative as a limit of a difference quotient, show that

$$a'(t) = a'(0)a(t).$$

2. Show that

$$a(t) = (1+i)^t$$

where $i = a(1) - a(0) = a(1) - 1$.

Solution

(a) Show that $a'(t) = a'(0)a(t)$

Start from the definition of the derivative:

$$a'(t) = \lim_{h \rightarrow 0} \frac{a(t+h) - a(t)}{h}.$$

Using the given property,

$$a(t+h) = a(t)a(h).$$

Substituting into the difference quotient gives

$$a'(t) = \lim_{h \rightarrow 0} \frac{a(t)a(h) - a(t)}{h}.$$

Factor out $a(t)$:

$$a'(t) = \lim_{h \rightarrow 0} \frac{a(t)(a(h) - 1)}{h} = a(t) \lim_{h \rightarrow 0} \frac{a(h) - 1}{h}.$$

Now compute $a'(0)$:

$$a'(0) = \lim_{h \rightarrow 0} \frac{a(0+h) - a(0)}{h}.$$

Since $a(0) = 1$ for an accumulation function,

$$a'(0) = \lim_{h \rightarrow 0} \frac{a(h) - 1}{h}.$$

Therefore,

$$\boxed{a'(t) = a(t)a'(0)}.$$

(b) Show that $a(t) = (1 + i)^t$

We know that

$$i = a(1) - a(0) = a(1) - 1.$$

Hence,

$$a(1) = 1 + i.$$

Now evaluate the functional equation repeatedly:

$$a(2) = a(1 + 1) = a(1)a(1) = (1 + i)^2,$$

and in general,

$$a(n) = a(1)^n = (1 + i)^n \quad \text{for } n \in \mathbb{Z}^+.$$

Because the relation

$$a(s + t) = a(s)a(t)$$

characterizes exponential behavior, the accumulation function has the form

$$\boxed{a(t) = (1 + i)^t}.$$

So the graph of $a(t)$ is an increasing exponential curve passing through $(0, 1)$ and taking the value $a(1) = 1 + i$ at $t = 1$.

